Data Analytics Capstone

Predicting Late Delivery Risk in Global E Commerce Fulfillment Using Machine Learning

Interim Report

Pankaj Nagpal

Walsh College

QM640: Data Analytics Capstone

Group 7

Sharath Srivatsa

Term 3

October 27, 2026

Project Repository and Data Access

GitHub: https://github.com/nagpalpankaj21/qm640-late-delivery-capstone

Publisher: https://data.mendeley.com/datasets/8gx2fvg2k6/5

Dataset DOI: https://doi.org/10.17632/8gx2fvg2k6.5

The public repository includes the executed interim analysis, cleaned order-level feature data, model outputs, and source-access instructions. The principal evaluator links were checked without sign-in on October 27, 2026. The analysis snapshot is commit 1bda9d8. Appendix A documents execution; Appendix B provides source and repository evidence.

Analysis folder:
https://github.com/nagpalpankaj21/qm640-late-delivery-capstone/tree/main/interim_2026_10_27

Report length and organization

The main narrative comprises 20 pages of double-spaced text. Exhibits, the title and navigation pages, references, and appendices are additional pages. Figures and tables are cited in the narrative. The report is self-contained and does not require the synopsis.

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Introduction

Background and Context

Global e-commerce fulfillment links order preparation, service promises, transportation, and customer communication. Late shipments increase exception handling and can weaken service credibility. Supply chain resilience research emphasizes visibility and coordinated responses to disruption (Christopher & Peck, 2004; Tang, 2006).

Fulfillment planners, logistics managers, and customer-service teams need early warning while an order can still be reviewed before dispatch. A probability estimate may help prioritize limited attention. Its value depends on timely predictors and a feasible review queue, not only retrospective classification accuracy.

Problem Statement

The problem is to estimate the published late-delivery risk of an unseen order using information available at order placement and before dispatch. The dependent variable is DataCo Late_delivery_risk. Independent variables describe promised service, destination, customer segment, transaction type, timing, quantity, price, discount, and basket composition. Realized shipping duration and delivery status disclose the outcome and are excluded from predictors. The gap addressed is a reproducible, interpretable comparison that respects this timing boundary and tests incremental value beyond service-mode rules.

Purpose of the Study

This retrospective quantitative study estimates adjusted associations, predicts late-label probabilities, compares three model families, and explains segment differences. It provides decision-support evidence rather than a production system. The performance-interpretability trade-off motivates transparent benchmarks (Baryannis et al., 2019); prediction alone does not establish that an intervention prevents delays or saves money.

Introduction

Interim Project Status (Progress Snapshot)

- Completed: source verification; order aggregation; missingness, duplicate, range, and metadata audits; development exploratory data analysis (EDA); fixed partitions; two benchmarks; restricted and expanded logistic models; random forest; gradient boosting; adjusted association estimates; training-only variance pilot; clustered uncertainty; segment explanations; temporal and label-definition sensitivities.
- In progress: threshold policy, training-only tuning, explanation stability, and operational review policy.
- Pending: final model freeze, untouched-test evaluation, external validation, and operational recommendations.

Analytical Objectives

The objectives are to estimate adjusted associations, evaluate pre-dispatch probabilities, compare transparent and nonlinear learners, and explain segment behavior. Each objective has a measurable output and a defined remaining check. The following research questions retain the synopsis scope while using actual interim evidence. Stakeholder review is required to set operational capacity and costs; neither is inferred from historical labels. The available sample and accessible numerical tools make the bounded retrospective study feasible.

Scope and objectives

The unit of analysis is a distinct order, not an order line. The 180,519 source lines become 65,752 eligible orders. Scope covers structured DataCo records from January 2015 through January 2018, four shipping modes, and five markets. Weather, carrier identifiers, live routing, causal intervention effects, and production integration are excluded. Python provides a feasible environment for the dataset size and the selected tabular classifiers.

Research Question 1 (RQ1)

Which pre-dispatch characteristics are associated with adjusted late-delivery odds? H01 sets the operational coefficients to zero. Logistic odds ratios and a clustered Wald test assess association.

Research Question 2 (RQ2)

How well can pre-dispatch information predict late labels? H02 sets receiver operating characteristic area under the curve (ROC-AUC) at or below .50; probability quality and review workload assess usefulness.

Research Question 3 (RQ3)

Which candidate balances performance and interpretability? H03 sets mean Brier losses equal for logistic regression, random forest, and gradient boosting. Three paired comparisons control familywise alpha .05.

Research Question 4 (RQ4)

How do driver contributions vary across modes and markets? H04 sets segment means equal. Adjusted contrasts examine normalized grouped logistic contributions; these explain the fitted model rather than causal effects.

Literature survey

Literature Review Approach

This targeted narrative review uses the synopsis bibliography, publisher records, Crossref metadata, and OpenAlex-indexed abstracts. Keywords cover supply chain risk, late delivery, interpretable prediction, random forest, gradient boosting, calibration, and clustered observations. Inclusion requires a scholarly source directly supporting a research question or analytical decision. Broad promotional claims and unverified performance numbers are excluded. The selection is purposeful, not an exhaustive systematic search.

Supply Chain Risk and Analytics

Christopher and Peck (2004) motivate resilience under supply chain vulnerability. Tang (2006) organizes risk-management perspectives. Ho et al. (2015) specifically review research from 2003 to 2013, categorizing risk definitions, types, factors, and mitigation strategies. These sources establish the operational problem and vocabulary, but their broad risk scope does not identify order-level late-label predictors or establish that changing a service category prevents delay.

Waller and Fawcett (2013) emphasize data-science definitions, domain knowledge, skills, and research opportunities. Choi et al. (2018) examine analytics techniques, computational challenges, and applications across forecasting, inventory, transportation, and risk. Unlike an order-level model experiment, these works support managerial integration and method selection. Ivanov et al. (2019) further connect digital technology with supply chain risk analytics; network-risk reasoning does not substitute for validation of this specific label.

Table 1 distinguishes each source setting, method, contribution, and RQ relevance. The synthesis motivates probability accuracy and review workload alongside ranking. It also separates conceptual relevance from empirical support: these papers justify studying the decision, while the present dataset must establish its own predictive evidence and operational limits.

Literature survey

Methods and Research Gap

Baryannis et al. (2019) examine the performance-interpretability trade-off in supply chain risk prediction. Toorajipour et al. (2021) systematically synthesize artificial intelligence across logistics, marketing, supply chain, and production. Yang et al. (2023) review 67 shortlisted articles from nine databases on machine learning and supply chain risk. These reviews map applications and gaps; their breadth does not establish which learner will work best on DataCo or transfer an effect size to this project.

Breiman (2001) and Friedman (2001) provide contrasting nonlinear approaches: forests average randomized trees, whereas boosting sequentially improves a loss objective. Their flexibility motivates interaction modelling, but adds complexity and tuning choices. Logistic regression supplies an interpretable additive comparator. This comparison requires common features and evaluation orders; otherwise algorithm differences can be confounded with different data or leakage controls.

Brier (1950) evaluates probability error, while Hanley and McNeil (1982) address AUC precision. DeLong et al. (1988) support paired treatment of correlated ROC comparisons. Efron (1979) motivates resampling; Liang and Zeger (1986) address correlated observations. These methods answer different questions, so ranking, probability quality, and clustered uncertainty are reported together. Lundberg and Lee (2017) support additive explanations of model behavior, not causal interventions.

The gap is an order-level, pre-dispatch DataCo evaluation that tests incremental value beyond service-mode rules while checking calibration, workload, and segment behavior. Broad reviews do not supply that comparison, and generic method papers cannot validate this outcome. The interim result sharpens the contribution: strong between-service separation coexists with weak within-mode ranking in several groups. Final work must establish useful individual-order prioritization, not merely reproduce service categories.

Literature survey

Directly Relevant Empirical Studies

Thomas and Panicker (2023) develop an order-delivery delay classifier using a public-repository dataset and report random-forest AUC .98. The publisher abstract verifies the task and score, but does not establish the detailed feature timing, aggregation, or validation protocol available behind the subscription. The result supports a forest comparator; it is not a directly comparable performance target and does not prove leakage or deployment usefulness.

İlbaş et al. (2026) study DataCo classification, delay-magnitude regression, and explanation. They describe a temporal split, report CatBoost classification AUC .867, and identify scheduled duration as the leading explanatory feature. Their source contains 180,519 lines and 65,752 orders, while the reported test population is 29,251. This differs from our 13,151 independent validation orders; a reconciled aggregation and eligibility protocol is needed before comparing scores.

The study combines operational, product, transaction, and geographic features and reports shipping-mode-specific patterns. However, its feature table includes profit and realized shipment/status fields without a fully explicit retained-input timing inventory. These fields require availability checks; the current capstone excludes realized information and uncertain profit timing. This is a transparency limitation, not a claim that the published model necessarily used leaked inputs.

Together, these empirical studies motivate nonlinear comparators, temporal evaluation, and segment explanations. They also show why prediction timing, analytical unit, feature eligibility, and sample denominators must accompany headline scores. Their stronger reported discrimination cannot be transferred to this capstone. Our cancellation-aware label audit and service-mode benchmark directly address those comparability and construct-validity concerns.

Table 1

Literature relevance matrix

Author (year)	Domain/context	Dataset/setting	Method	Key contribution	Project/RQ relevance
Christopher & Peck (2004)	Supply chain resilience	Conceptual; no DataCo sample	Framework	Visibility and collaboration support resilience	Frames stakeholders and early review; RQ2/RQ4
Tang (2006)	Supply chain risk	Prior risk literature	Review	Organizes risk and mitigation perspectives	Bounds operational claims; all RQs
Ho et al. (2015)	Risk management	Research from 2003-2013	Literature synthesis	Categorizes risk definitions, factors, and mitigation	Frames scope; does not validate this outcome; RQ1/RQ4
Waller & Fawcett (2013)	Supply chain analytics	Managerial research agenda	Conceptual synthesis	Links domain knowledge and analytical skills with supply chain management	Supports adoption and workload context; RQ2

Table 1 continued

Literature relevance matrix

Author (year)	Domain/context	Dataset/setting	Method	Key contribution	Project/RQ relevance
Choi et al. (2018)	Operations analytics	Operations literature and enterprise cases	Techniques and applications review	Compares capabilities and computational/data challenges	Supports feasible tabular approach; RQ2
Baryannis et al. (2019)	Supply chain risk prediction	Applied prediction setting	Machine-learning comparison	Examines performance-interpretability trade-off	Motivates transparent versus flexible models; RQ3
Toorajipour et al. (2021)	Artificial intelligence	Logistics, marketing, supply chain, production	Systematic review	Maps current/potential techniques and application gaps	Supports bounded scope; no DataCo performance claim; RQ2/RQ3
Yang et al. (2023)	Machine learning risk	67 articles; 9 databases	Literature review	Maps machine-learning risk literature and research gaps	Motivates evaluation; review scope limits transfer; RQ2/RQ3

Table 1 continued

Literature relevance matrix

Author (year)	Domain/context	Dataset/setting	Method	Key contribution	Project/RQ relevance
Breiman (2001)	Tabular learning	Methodological experiments	Randomized tree ensemble	Introduces random forests	Nonlinear interaction comparator; RQ2/RQ3
Friedman (2001)	Function estimation	Methodological experiments	Gradient boosting	Develops sequential loss minimization	Flexible probability comparator; RQ2/RQ3
Thomas & Panicker (2023)	Order-delivery delay	Public-repository data; abstract	Classifier comparison	Reports forest AUC .98; detailed protocol not verified	Forest comparator; no direct score transfer; RQ2/RQ3
İlbaş et al. (2026)	DataCo delay and explanation	180,519 lines; test n=29,251	Temporal ensemble study	Reports CatBoost AUC .867; scheduled duration dominates	Compare timing, unit and population; RQ2-RQ4

Note. Conceptual and methodological evidence is distinguished from findings on the current dataset.

Data Description

Data Source(s) and Access

DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS, Version 5 (Constante et al., 2019), is available at https://data.mendeley.com/datasets/8gx2fvg2k6/5. The structured CSV supplies records; the description CSV supplies metadata. Clickstream data are excluded. CC BY 4.0 requires attribution and disclosure of transformations. Appendices A and B document reproduction and source evidence.

Dataset Overview

The verified source contains 180,519 line items, 53 fields, and 65,752 orders dated January 1, 2015 through January 31, 2018. There are 36,048 positive and 29,704 non-positive orders (54.82% positive). This publisher late-sending indicator is not an independently verified arrival-delay measurement. Development labels equal duration above promise for non-canceled shipments; canceled shipments are non-late (Table 6).

The frozen split contains 39,450 training, 13,151 validation, and 13,151 reserved-test orders. EDA uses 52,601 development orders. Table 2 records analytical definitions, types, units, roles, and handling; Figure 1 previews non-identifying feature records. The full 53-field source dictionary is accessible in the repository.

GitHub Data Availability Statement

Verified public snapshot: 1bda9d8. The interim_2026_10_27/ folder contains processed_orders.csv, four execution scripts, dependencies, and results. The data/raw/README.md provides exact source filenames and download instructions. Raw identifying records remain with the publisher; the public feature export omits original identifiers, addresses, passwords, and placement timestamps.

Table 2

Analytical data dictionary

Variable	Definition	Data type	Values/units	Missing handling	Role and relevance
Late_delivery_risk	Published late-sending label	Binary	0 or 1	No missing; never impute	Target in all RQs; not an input
Shipping Mode	Promised service category	Nominal	4 categories	No missing	Predictor; RQ4 segment; pre-dispatch
Scheduled days	Promised shipping duration	Integer	0, 1, 2, 4 days	No missing	Audit only; redundant with mode
Market; Order Region	Broad destination; region	Nominal	5 markets; source regions	No missing	Predictors; market is RQ4 segment
Customer Segment	Customer business segment	Nominal	3 source categories	No missing	Predictor; no customer identifier
Type	Transaction/payment type	Nominal	4 source categories	No missing	Predictor; timing requires operational confirmation
order_quantity	Sum of line quantities	Integer	Positive units/order	No missing	Engineered predictor; standardize in training
weighted_unit_price	Sum(quantity x price)/total quantity	Continuous	Positive source currency units	No missing	Engineered predictor; currency unverified
discount_rate_mean	Sum(quantity x discount)/total quantity	Continuous	Proportion 0 to 1	No missing	Engineered predictor; observed 0 to .25

Table 2 continued

Analytical data dictionary

Variable	Definition	Data type	Values/units	Missing handling	Role and relevance
order_month	Month of placement	Categorical	1 to 12	Parse succeeds	Engineered predictor; nominal encoding
order_weekday	Weekday of placement	Categorical	0 to 6; Monday = 0	Parse succeeds	Engineered predictor; nominal encoding
dominant_category	Category with greatest summed quantity	Nominal	Source category labels	No missing	Engineered predictor; alphabetical ties
dominant_department	Department with greatest summed quantity	Nominal	Source department labels	No missing	Constructed; omitted from expanded models
mixed_category	Order includes multiple categories	Binary	0 or 1	No missing	Engineered predictor; basket complexity
Order Id; placement date	Private grouping key; timestamp	Identifier; datetime	Source codes; 2015-2018	No missing	Grouping/derivation only; not model inputs
Customer Id	Private dependence key	Identifier	Source customer codes	No missing	Cluster inference only; never public input
Delivery Status	Realized shipment category	Nominal	Source status labels	No missing	Eligibility sensitivity only; excluded predictor

Figure 1

Rendered preview of non-identifying processed training records

Shipping mode	Market	Quantity	Mixed	Late label
Standard Class	LATAM	10	1	1
Standard Class	LATAM	6	1	1
Standard Class	LATAM	18	1	0
Standard Class	LATAM	4	1	1
Standard Class	LATAM	9	1	1
Standard Class	LATAM	7	0	0

Note. Actual processed records; original order/customer identifiers and addresses are omitted. Mixed = multiple categories; late label is the binary target. This is a rendered data preview, not a source-system screen capture.

Current public repository structure

qm640-late-delivery-capstone/

README.md; requirements.txt; .gitignore

data/raw/README.md; data/raw_data_dictionary.csv

scripts/profile_data.py; outputs/source_profile.json

interim_2026_10_27/README.md; processed_orders.csv

interim_2026_10_27/baseline_audit.py; enhance_analysis.py

interim_2026_10_27/pilot_planning.py; enhanced_results.json

Public files verified without sign-in on October 27, 2026. The complete repository-tree snapshot is shown in Appendix B.

Analysis

Data Cleaning

The structured file is read with Latin-1 encoding and checked against the source SHA-256 before transformation. There are zero full-row duplicates and zero duplicated Order Item Id values. Repeated Order Id values represent legitimate basket lines. Deleting them as duplicate records would discard products and distort the analytical unit. Instead, quantities are summed and monetary characteristics are quantity-weighted.

Eight order metadata fields are checked for within-order consistency: target, shipping mode, market, region, customer segment, transaction type, scheduled duration, and placement timestamp. All show zero conflicts. Quantities and unit prices must be positive and finite; discounts must be finite and within zero to one. Explicit date parsing and all numeric eligibility checks pass. No analytical input is missing, so no predictor imputation or eligibility exclusion is needed.

Four excluded source fields contain nulls: surname has 8, customer ZIP has 3, destination ZIP has 155,679, and product description has 180,519. Their respective missingness rates are 0.0044%, 0.0017%, 86.24%, and 100%. These fields are not imputed because they are outside the feature specification. Zero nulls elsewhere do not establish valid timing, measurement accuracy, or absence of hidden placeholder values.

Training-derived interquartile-range (IQR) fences flag 75 quantity and 5,196 weighted-price observations in development, or 0.14% and 9.88%; discount flags are zero. Valid extreme values are retained because a statistical flag does not establish recording error. Table 3 records detection, treatment, and rationale. Quantity-weighted category totals select dominant category and department with alphabetical tie-breaking. Missing targets or grouping keys would require exclusion in a new source version, never target imputation.

Table 3

Executed cleaning and aggregation log

Issue	Detection	Treatment	Count / percentage	Rationale
Missing source fields	Null count	Exclude four unrelated fields	8 surname; 3 customer ZIP; 155,679 order ZIP; 180,519 description	Identifying, highly incomplete, or unstructured; no imputation
Analytical missingness	Null and parse audit	No imputation	0 input gaps; 0 failed dates	Retain valid observed inputs
Duplicates	Full rows; item identifiers	No deletion	0; 0	Repeated order keys are basket lines
Metadata conflicts	Within-order unique values	Aggregate after audit	0 in each of 8 fields	One consistent target and profile
Numeric invalidity	Finite and allowed range	No exclusion	0 invalid lines; 0 excluded orders	Positive quantities/prices; discounts 0-1
Statistical extremes	Training 1.5-IQR fences	Retain valid values	75 quantity (0.14%); 5,196 price (9.88%); 0 discount	Flags do not prove errors
Order aggregation	Group by private Order Id	Sum / weighted means	180,519 lines to 65,752 orders	Avoid repeated-label pseudoreplication

Analysis

EDA Results for Service and Geography

Development prevalence is 54.82%, so automatic oversampling is unnecessary. A rule that flags every order achieves full recall while reviewing the entire population. This illustrates why accuracy or recall alone cannot establish operational usefulness. Figure 2 and Table 4 instead examine segment rates, denominators, and the variables that may drive useful prioritization.

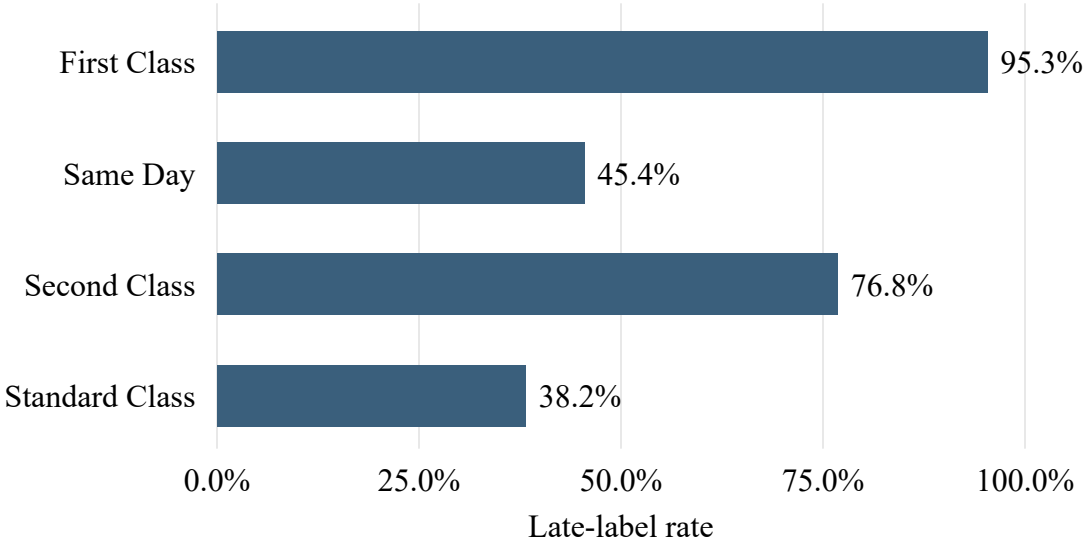
First Class has 7,681 positive labels among 8,060 development orders, or 95.30%. Second Class has 7,848 of 10,219, or 76.80%. Same Day has 1,287 of 2,832, or 45.44%. Standard Class has 12,022 of 31,490, or 38.18%. The approximately 57-percentage-point First-versus-Standard difference is large enough to make a training-fitted shipping-mode lookup an essential comparator.

Scheduled durations correspond exactly to the four mode categories in this source. Including both would duplicate the same service information, making individual regression effects difficult to identify. Mode is retained and scheduled days excluded from the model matrix. The observed relationship may reflect tighter promises, label construction, omitted processes, or genuine execution differences; it cannot demonstrate that faster service causes delay.

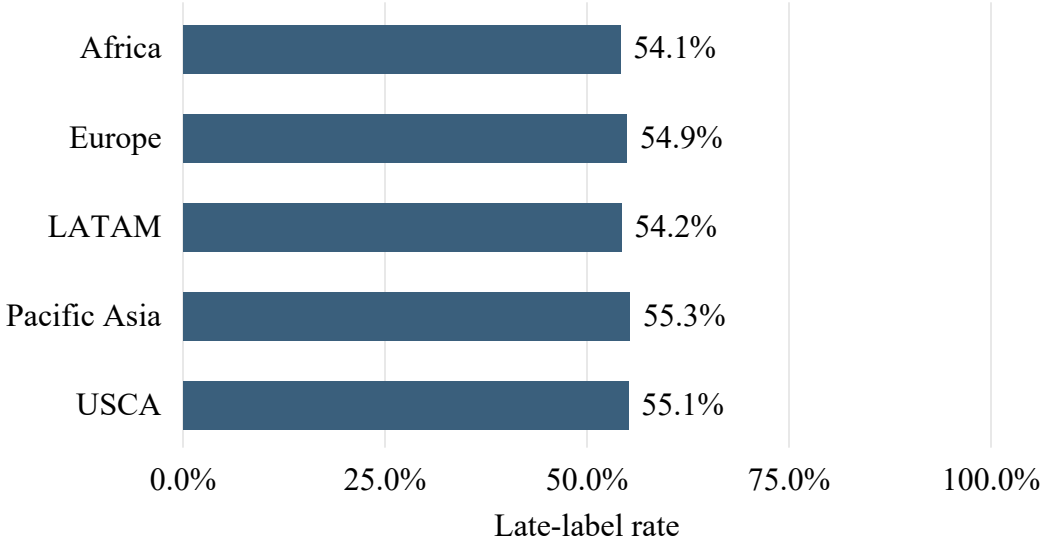
Broad-market rates lie between 54.09% in Africa and 55.29% in Pacific Asia. Consumer, Corporate, and Home Office rates are also close, at 54.67%, 54.99%, and 54.98%. Small raw differences do not establish equivalence, but they give little support for major geographic or customer-segment claims without adjustment. For RQ1 the key descriptive priority is service mode; for RQ4 the next question is performance within each service category, rather than only the pooled rate.

Figure 2

Development late-label rates by service and market



Note. Service counts: First Class 8,060; Same Day 2,832; Second Class 10,219; Standard Class 31,490. All rates describe the publisher label.



Market counts: Africa 3,078; Europe 14,978; LATAM 13,661; Pacific Asia 14,058; USCA 6,826. Common zero-to-one axes avoid exaggerating market differences.

Analysis

EDA Results for Time and Order Composition

Development orders average 5.85 units, with median 5 and interquartile range 2 to 9. Weighted unit price averages 135.12 source currency units, compared with median 92.21 and interquartile range 59.99 to 149.99. The price maximum is 1,500, consistent with a pronounced right tail. The currency denomination is not established by the supplied metadata and is not assumed to be US dollars.

Weighted discount averages 0.1018, with median 0.10 and observed range zero to 0.25. All values are within the admissible proportion range. Multi-category orders account for 35,624 of 52,601 development observations, or 67.72%. Dominant category and a mixed-category flag therefore describe basket composition without treating individual product lines as independent outcomes. Their operational relevance is plausible, but their incremental prediction value must be measured.

Annual late-label rates are 54.46% in 2015, 55.26% in 2016, 54.55% in 2017, and 56.94% in January 2018. The final period contains only 1,700 development orders and is not comparable with a complete year. Table 4 summarizes numeric distributions; Figure 3 shows annual rates with denominators. These summaries suggest relative stability at a broad level, not proof of future-period transportability.

EDA leads to four analytical decisions: retain valid extreme prices; represent basket composition explicitly; avoid the duplicate service-duration input; and conduct a chronological sensitivity check. It also motivates the feature-removal experiment. If predictive ranking collapses when service mode is removed, the model mainly summarizes service distinctions rather than uncovering substantial additional order-specific information. These decisions link the descriptive work directly to RQ1, RQ2, and RQ3. Table 5 consolidates the EDA insights and next decisions; Table 6 audits the outcome definition.

Table 4

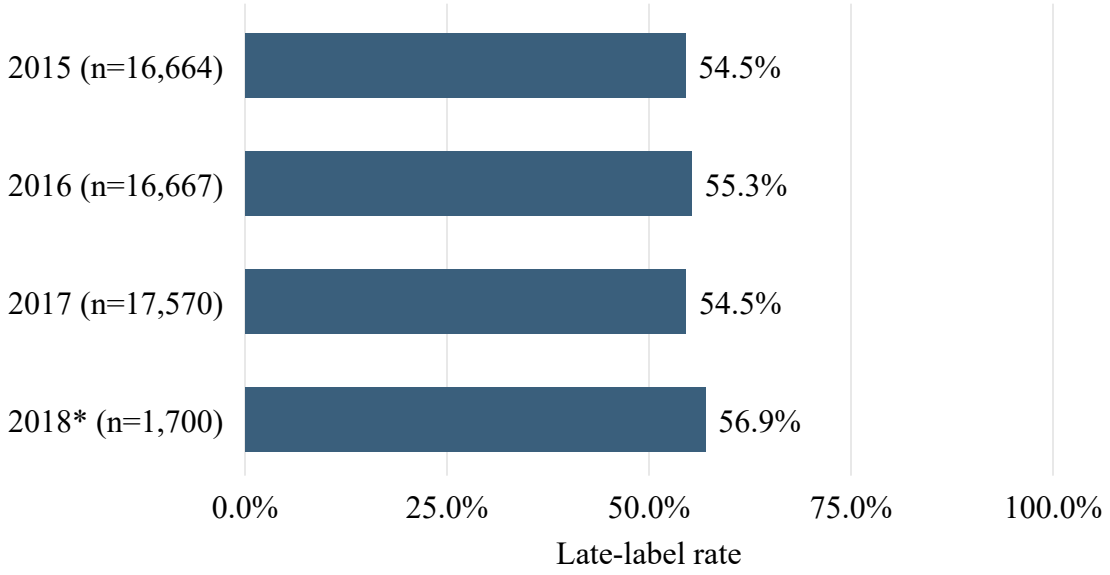
Development numeric summaries

Feature	Mean	Median	Q1 to Q3	Range
Quantity	5.85	5.00	2 to 9	1 to 24
Weighted price	135.12	92.21	59.99 to 149.99	9.99 to 1,500
Weighted discount	0.1018	0.10	0.06 to 0.14	0 to 0.25

Note. Development n = 52,601; prices are unverified source currency units.

Figure 3

Annual development late-label rate



Note. *2018 includes January only. The partial period is not comparable to a full calendar year. Numeric distributions are right-skewed, particularly weighted price.

Table 5

EDA insight summary

Reference	What it shows	Key insight	RQ/objective link	Decision / next step
Figure 2	Mode late-label rates	First 95.30%; Standard 38.18%	RQ1/RQ2: large service separation	Retain mode lookup; avoid causal claims
Figure 2	Broad market rates	54.09%-55.29%: small spread	RQ1/RQ4: raw rates are not adjusted effects	Check adjusted and within-segment behavior
Table 4	Numeric distributions	Price skew; valid extreme values	RQ2: trimming may remove real products	Retain valid values; train-only transformations
Figure 3	Annual rates and counts	2018 contains January only	RQ2/RQ3: apparent trend may reflect period mix	Use chronological sensitivity
Basket audit	35,624 mixed-category orders	67.72% of development baskets mix categories	RQ2: plausible complexity, not proven benefit	Compare engineered features against baseline
Table 6	Duration/status reconciliation	Canceled orders explain 1,306 simple-rule exceptions	All RQs: outcome is promise-relative shipping risk	Keep realized fields out of predictors

Note. Development n = 52,601. Each descriptive finding supports an analytical decision; these are not final-test or causal conclusions.

Table 6

Development outcome-label reconciliation

Source status	Orders	Late label = 1	Real days > scheduled	Simple-rule exceptions
Advance shipping	12,078	0	0	0
Late delivery	28,838	28,838	28,838	0
Shipping canceled	2,281	0	1,306	1,306
Shipping on time	9,404	0	0	0
Total	52,601	28,838	30,144	1,306

Note. The simple rule labels an order late whenever actual shipping days exceed scheduled days. It agrees with 51,295 orders (97.52%). Every exception is canceled. The cancellation-aware rule agrees with all 52,601 development labels; its exception count is zero in every status group.

Interpretation

On the development population, Late_delivery_risk equals 1 exactly when actual shipping days exceed scheduled days and status is not Shipping canceled. The 2,281 canceled orders all have label 0; 1,306 nevertheless exceed the scheduled duration. Non-canceled orders have no disagreement with the simple duration rule. This reconciles the label with its shipping fields but does not independently validate customer arrival dates.

Rule agreement is a data-definition check, not classifier performance. Actual duration and realized status remain excluded from predictive inputs. The result explains why service promises and cancellation treatment matter when interpreting the strong shipping-mode signal. Reserved test orders are excluded from this reconciliation.

Analysis

Sample Size for RQ1 and RQ2

Sample-size planning follows the estimand for each question. RQ1 and RQ2 retain confidence interval (CI) precision methods; RQ3 and RQ4 now use power-based comparisons. Alpha is 0.05, desired power is 0.80 where applicable, and the planned development/test allocation is 0.80/0.20. Parameters are planning assumptions, not retrospective claims about achieved power. Table 7 records every step and the allocation conversion.

RQ1 targets a 95% interval half-width h = 0.20 for the log-odds coefficient of Standard Class versus other services. Assume outcome probability p = 0.50, exposure variance 0.25, and variance-inflation component R-squared = 0.25. Per-order information is I = p(1-p)Var(X)(1-R-squared) = 0.50 x 0.50 x 0.25 x 0.75 = 0.046875. Required development n = ceiling[1.96 squared / (0.20 squared x 0.046875)] = ceiling(2,048.85) = 2,049. Total N = ceiling(2,049/0.80) = 2,562.

RQ2 targets AUC A = 0.70 with 95% half-width e = 0.03 and balanced planning classes. Hanley and McNeil (1982) give Q1 = A/(2-A) = 0.538462 and Q2 = 2A squared/(1+A) = 0.576471. The balanced variance simplifies to 0.269864/n + 0.300271/n squared. Setting 1.96 squared times this variance equal to 0.03 squared yields n approximately 1,153.012. Round upward to 1,154 test orders, or 577 in each class; total N = 1,154/0.20 = 5,770.

The RQ1 approximation concerns one contrast, not every rare category. RQ2 planning assumes independent orders and an approximate AUC; final uncertainty uses observed predictions and clustered resampling.

Analysis

Sample Size for RQ3 and RQ4

RQ3 compares paired Brier losses for three model pairs. Two-sided alpha per pair is .05/3, z-alpha = 2.39398, and power .80 gives z-power = .84162. With planned absolute loss difference .01 and paired standard deviation (SD) .20, required test n = ceiling[((2.39398+.84162).20/.01) squared] = 4,188. Total N = 4,188/.20 = 20,940. Table 7 shows the intermediate algebra.

RQ4 compares normalized principal-driver means. Up to ten pairwise contrasts within a segment family give alpha = .005 and z-alpha = 2.80703. With standard deviation .20, difference .05, and power .80, n per segment = ceiling[2((2.80703+.84162).20/.05) squared] = 427. At minimum segment share .05 and test allocation .20, total N = 427/.01 = 42,700.

A three-fold training-only pilot (n = 39,450; seed 644) supports retaining SD = .20. Out-of-fold paired-loss SDs are .0366-.0449; segment-contribution SDs are .0375-.0979. Thus .20 exceeds the observed maxima by about 4.45 and 2.04 times. The .01 Brier target is about 4% of a .25 reference loss; .05 denotes five percentage points of attribution share. These are planning thresholds, not validated business benefits.

Final N = max(2,562; 5,770; 20,940; 42,700) = 42,700 independent-equivalent orders; 65,752 are available. A 1.5 design-effect allowance requires 64,050. At SD .30, RQ4 requires 95,900 and would be infeasible. Pilot variability does not guarantee future power; customer dependence and final segment counts still require checks.

Table 7

Sample-size methods and stepwise calculations

RQ	Method / assumptions	Intermediate calculation	Required analysis n	Total N
RQ1	95% coefficient CI; h=.20; p=.50; Var(X)=.25; R-squared=.25	I=.5 x .5 x .25 x .75=.046875; 3.8416/(.04 x .046875)=2,048.85	2,049 development	ceil(2,049/.8) = 2,562
RQ2	95% AUC CI; A=.70; e=.03; balanced classes	Q1=.538462; Q2=.576471; solve 3.8416(.269864/n + .300271/n^2)=.0009; n=1,153.012	1,154 test (577/class)	1,154/.2 = 5,770
RQ3	Paired Brier power; alpha=.05/3; power=.80; delta=.01; SD=.20	z-alpha=2.39398; z-power=.84162; [(z-alpha+z-power).20/.01]^2=4,187.6456	4,188 test	4,188/.2 = 20,940
RQ4	Two-mean power; alpha=.05/10; power=.80; delta=.05; SD=.20	z-alpha=2.80703; 2[(z-alpha+.84162).20/.05]^2=426.0059	427 per segment	427/(.05 x .2) = 42,700

Note. Final N = max(2,562; 5,770; 20,940; 42,700) = 42,700. Retained N = 65,752. Full-precision critical values precede upward rounding. A 3-fold training-only pilot (39,450 orders; seed 644) gives maximum paired-loss SD .04494 and maximum segment SD .09792, below the retained .20. A 1.5 design-effect allowance requires 64,050. At SD=.30, RQ4 needs 95,900; adequacy remains assumption-dependent.

Modelling

Choice of Models with Justification

The modelling problem is binary classification with probability outputs. A constant benchmark assigns training prevalence to every order. A second benchmark maps shipping mode to its training late-label rate. These require no validation outcomes when scoring and establish whether additional modelling improves on the strongest operational distinction. The previously submitted restricted logistic baseline is reproduced before extending the feature set.

The expanded logistic model estimates additive log odds with L2 regularization, C = 1, and a maximum of 2,000 iterations. It offers transparent source-feature contributions and efficient scoring. Its limitations include additive functional form and dependence on coding choices. Breiman's (2001) random forest provides nonlinear interactions through 240 trees, maximum depth 14, minimum leaf size 30, and square-root feature subsampling. These controls limit overly specific leaves in a substantial but historical dataset.

Histogram gradient boosting follows the sequential learning principle of Friedman (2001). The fixed recipe uses 180 iterations, learning rate 0.07, 15 leaves, minimum leaf size 40, and L2 regularization 5. Early stopping is disabled so validation labels do not control training iterations. These are prespecified initial configurations for this extension, not the outcome of an exhaustive hyperparameter search.

All three expanded candidates receive the same features, transformations, training orders, and validation orders. Table 8 records the feature specification and Table 10 compares performance. The separate RQ1 model is unpenalized, uses the prespecified binary service contrast, and reports adjusted odds ratios with customer-cluster uncertainty. Its coefficients answer an association question; predictive penalized coefficients are not substituted for inferential tests. More elaborate tuning and calibration remain development tasks before final evaluation.

Modelling

Features Included and Feature Engineering

Expanded predictive inputs are order quantity, quantity-weighted price and discount, mixed-category status, shipping mode, market, customer segment, transaction type, month, weekday, destination region, and dominant category. Dominant department is constructed for documentation but omitted from modelling to reduce redundant product hierarchy. Scheduled days are omitted because they duplicate shipping mode. Table 8 identifies original and engineered features and their model usage.

Numeric standardization and categorical one-hot encoding are fitted on the 39,450 training orders only. Unseen categories are handled by ignoring their indicator contribution. The expanded matrix has 112 encoded features. Keeping both market and region is acceptable for regularized prediction, but nested geography complicates individual coefficient interpretation. The inferential specification therefore uses market without region and does not claim separate effects for every nested product or destination level.

Validation and Leakage Controls

All aggregation occurs within an order before splitting. No order can appear in multiple partitions. Final test orders are excluded from EDA, model fitting, validation comparisons, and robustness evaluation. Private customer identifiers support clustered uncertainty but are absent from model inputs and public feature exports. Realized duration, delivery status, dispatch timestamp, downstream order status, and fields with uncertain availability are excluded.

Validation is visible to development. Three-fold training cross-validation estimates planning variance only; no broad model tuning is claimed. Customer overlap between training and validation still limits new-customer generalization, even when intervals account for repeated customers in evaluation. A chronological sensitivity uses only development orders, fitting before 2017 and evaluating from January 2017 onward. A later customer-separated assessment remains necessary for claims about previously unseen customers.

Table 8

Feature set and analytical usage

Feature	Original/engineered	Type / transformation	Reason	Usage
Shipping Mode	Original	Nominal; one-hot	Service promise; strongest descriptive signal	All prediction models; binary contrast for RQ1
Market; Customer Segment; Type	Original	Nominal; one-hot	Broad destination and order context	Restricted and expanded prediction; RQ1 adjustment
Order Region	Original	Nominal; one-hot	Finer destination context	Expanded prediction only
Quantity; weighted price; weighted discount	Engineered order aggregates	Numeric; training standardization	Basket size and monetary characteristics	All logistic/tree prediction; RQ1 increments specified
Month; weekday	Engineered from placement	Categorical; one-hot	Calendar variation	All fitted prediction models; RQ1 adjustment
Dominant category; mixed-category flag	Engineered basket mix	Nominal / binary	Product composition without line duplication	Expanded prediction; grouped explanation
Scheduled days; dominant department	Original / engineered	Audit only	Redundant service / product hierarchy	Omitted from predictive feature matrix
IDs; realized status/duration; addresses	Original	Excluded	Memorization, privacy, or outcome leakage	IDs for private grouping; status for sensitivity only

Note. Expanded encoded columns = 112. One-hot encoding includes all levels for regularized prediction; unseen levels contribute zero indicators. The RQ1 model uses reference coding and 31 coefficients including intercept.

Modelling

Evaluation Metrics (Include Formulae and Calculations)

Let TP and TN denote correctly classified late and non-late orders; FP denotes a false late alert and FN a missed late order. At probability threshold 0.50, the expanded logistic model has TP = 3,941, TN = 5,206, FP = 735, and FN = 3,269. Table 12 displays this confusion matrix. The total is 13,151. Accuracy = (TP+TN)/n = 9,147/13,151 = 0.6955. Precision = TP/(TP+FP) = 3,941/4,676 = 0.8428. Recall = TP/(TP+FN) = 3,941/7,210 = 0.5466.

Specificity = TN/(TN+FP) = 5,206/5,941 = 0.8763. F1 = 2TP/(2TP+FP+FN) = 7,882/11,886 = 0.6631. Alert rate = (TP+FP)/n = 4,676/13,151 = 0.3556. These are actual validation counts, not illustrative data. The threshold is a common reference, not an optimized operational policy.

Brier score = mean[(predicted probability - observed outcome) squared] = 0.19509; lower is better (Brier, 1950). ROC-AUC measures positive-negative ranking with half credit for ties and is 0.73964. Average precision sums threshold precision weighted by recall increments and is 0.81396; it differs from trapezoidal precision-recall area. The no-skill precision reference is the validation prevalence, 0.54825. Table 9 provides definitions. Figure 4 reports calibration and expected calibration error (ECE).

The 95% AUC and Brier intervals use 500 customer-cluster resamples (seed 643). The primary paired Brier comparisons now use 5,000 whole-customer resamples (seed 645), with 98.333% percentile intervals for three Bonferroni-adjusted pairs. There are 9,151 validation customers. Approximately 42 draws inform each adjusted tail, versus four with 500 replicates. Models remain fixed, so these intervals exclude refitting and model-search uncertainty. Order-bootstrap results remain a sensitivity check.

Table 9

Evaluation metrics and formulae

Metric	Formula / definition	Interpretation
Accuracy	(TP + TN)/n	Correct classifications
Precision	TP/(TP + FP)	Positive yield among alerts
Recall	TP/(TP + FN)	Share of late labels captured
Specificity	TN/(TN + FP)	Share of non-late labels unflagged
F1	2TP/(2TP + FP + FN)	Precision-recall balance
ROC-AUC	P(score+ > score-) + .5 P(tie)	Positive-negative ranking
Average precision	Sum precision x recall increment	Precision-recall ranking; prevalence reference
Brier	mean[(p-y)^2]	Probability error; lower is better
Log loss	-mean[y ln p + (1-y) ln(1-p)]	Probability penalty; lower is better
ECE	Sum (n-bin/n) x |mean(p)-mean(y)|	Calibration error; depends on bins
Alert rate	(TP + FP)/n	Review workload at given threshold

Note. Classification uses threshold .50. AUC/AP have no single threshold. Probability quality and calibration are reported alongside classification results.

Preliminary results

Preliminary Findings for RQ1 and RQ2

RQ1: the adjusted Standard Class odds ratio is 0.155, with customer-cluster 95% interval 0.149 to 0.162 and p below .001. This corresponds to approximately 84.5% lower modeled odds relative to the pooled other services, conditional on the specified covariates. It is not an 84.5% reduction in probability and is not a causal effect. The broad null that all non-intercept coefficients are zero is rejected by the clustered Wald test; Table 11 reports selected estimates.

Quantity per five units has odds ratio 0.997, price per 100 source currency units 1.004, and discount per 0.10 proportion 0.997. Their intervals include one. These results provide no clear adjusted association for those increments under the current model, but do not prove that all functional forms or interactions are absent. The primary service contrast pools heterogeneous other modes and requires a four-level sensitivity specification in final work.

RQ2: the expanded logistic model attains AUC 0.7396, average precision 0.8140, and Brier 0.19509. Its customer-bootstrap AUC interval is approximately 0.7311 to 0.7471, above the chance reference. The earlier restricted baseline remains slightly better in AUC and Brier; adding product and regional fields has not established useful incremental value. The mode lookup already reaches AUC 0.7295 and Brier 0.19534.

Removing shipping mode from the expanded logistic recipe yields AUC 0.5365 and Brier 0.24641, close to the constant benchmark. The result is a controlled feature-removal comparison, not a causal estimate of shipping service. It shows that overall ranking mostly depends on service information. Final RQ2 confirmation requires the frozen model, threshold, and feature choices to be evaluated on the reserved test set.

Table 10

Preliminary validation model performance

Model	AUC	Avg. precision	Brier	Recall	Precision
Constant	0.5000	0.5482	0.24767	1.0000	0.5482
Mode lookup	0.7295	0.7505	0.19534	0.5401	0.8526
Logistic restricted	0.7412	0.8153	0.19457	0.5433	0.8484
Logistic full	0.7396	0.8140	0.19509	0.5466	0.8428
Random forest	0.7392	0.8146	0.19523	0.5718	0.8198
Gradient boosting	0.7373	0.8137	0.19464	0.5469	0.8438

Note. Common validation n = 13,151; positives = 7,210. All classifications use .50. Expanded recipes share features and split; the restricted baseline has fewer features. This is preliminary validation, not final-test performance.

Table 11

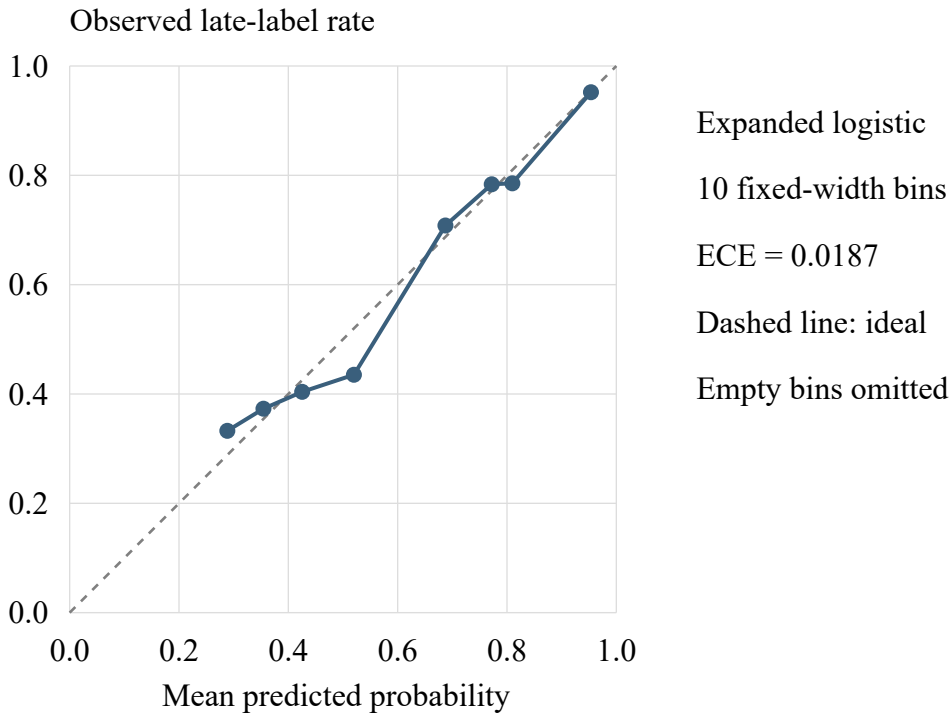
Selected adjusted logistic associations

Contrast / increment	Odds ratio	Clustered 95% CI	p
Standard vs others	0.1551	0.1489 to 0.1615	< .001
Quantity +5 units	0.9968	0.9726 to 1.0217	.801
Price +100 units	1.0044	0.9903 to 1.0186	.545
Discount +.10	0.9967	0.9626 to 1.0319	.850

Note. Development n = 52,601; customer clusters = 18,841. Adjusted for market, segment, transaction type, month and weekday. Wald statistic = 8287.89, df=30, $p < .001$. The primary contrast is an adjusted association, not a causal effect.

Figure 4

Expanded logistic calibration on validation orders



Note. Fixed-width probability bins; point positions use observed bin averages. ECE is descriptive and does not ensure calibration within every segment.

Table 12

Expanded logistic confusion matrix

Actual label	Predicted non-late	Predicted late
Non-late	5,206 (TN)	735 (FP)
Late	3,269 (FN)	3,941 (TP)

Note. Threshold .50; 4,676 alerts (35.56%). Missed positive labels = 45.34%.

Preliminary results

Preliminary Findings for RQ3

The three expanded candidates yield similar validation performance. Logistic regression has AUC 0.7396 and Brier 0.19509; random forest has AUC 0.7392 and Brier 0.19523; gradient boosting has AUC 0.7373 and Brier 0.19464. The restricted logistic baseline retains the smallest reported Brier, 0.19457. Thus gradient boosting is the best expanded candidate on probability error, while the simpler restricted model remains a serious overall comparator.

The paired Brier differences are small. Expanded logistic minus forest is approximately -0.000143; logistic minus boosting is 0.000444; forest minus boosting is 0.000587. All three customer-bootstrap multiplicity-adjusted intervals include zero, as shown in Table 13. These exploratory comparisons do not establish reliable superiority. Failure to find a difference is also not evidence of equivalence; no equivalence margin or dedicated equivalence test was specified.

At threshold 0.50, forest recall rises to 0.5718 compared with 0.5466 for expanded logistic, but precision falls to 0.8198 and alert workload increases to 38.24%. A small recall improvement therefore carries additional reviews and false alerts. The preferred recipe cannot be chosen solely from one threshold-specific metric or from an insignificant ordering of near-equal Brier scores.

Table 14 summarizes robustness checks. Temporal sensitivity fits expanded logistic on 33,331 development orders before 2017 and evaluates 19,270 later development orders. AUC is 0.7396 and Brier 0.19616, close to the random validation result. Removing 568 canceled validation orders leaves 12,583 and gives AUC 0.7397 and Brier 0.19017. The denominator and prevalence change, so this is a label-population sensitivity, not proof of model improvement. Table 18 consolidates the four interim answers and remaining checks. Final tuning, calibration, and independent evaluation remain pending.

Table 13

Paired probability-error comparisons

Model A minus model B	Brier difference	Adjusted interval
Logistic full vs Random forest	-0.000143	-0.000970 to 0.000682
Logistic full vs Gradient boosting	0.000444	-0.000300 to 0.001183
Random forest vs Gradient boosting	0.000587	-0.000081 to 0.001202

Note. Customer-cluster bootstrap; 5,000 draws; seed 645; 98.333% intervals for three pairwise comparisons. Negative favors A. All intervals include zero. Conditional on fitted models; no equivalence claim.

Table 14

Expanded logistic robustness checks

Assessment	Evaluation n	AUC	Brier
Random validation	13,151	0.7396	0.19509
Without mode	13,151	0.5365	0.24641
Later development	19,270	0.7396	0.19616
Exclude canceled	12,583	0.7397	0.19017

Note. Temporal fit uses 33,331 development orders before 2017. Excluding canceled changes the evaluation population and prevalence; these rows are not head-to-head model improvements.

Preliminary results

Preliminary Findings for RQ4

Expanded logistic explanations are additive on the log-odds scale. Each encoded contribution is its coefficient times the feature value minus its training mean. One-hot contributions are grouped by source variable. The baseline plus grouped contributions reconstructs the score to numerical tolerance (Lundberg & Lee, 2017). These explain the logistic model only.

Figure 5 summarizes the grouped explanation ranking. Shipping mode has mean absolute contribution 1.122 log-odds units, followed by transaction type at 0.190 and dominant category at 0.044. RQ4 prespecifies shipping mode as the principal driver based on the earlier development evidence. Its normalized absolute share is the absolute grouped contribution divided by the sum across all groups; a zero denominator is assigned zero. Correlated inputs and encoding choices affect this attribution, and the independent-background interpretation may describe unrealistic feature combinations.

Mean normalized mode shares are .881 for First Class, .628 for Same Day, .695 for Second Class, and .714 for Standard Class. Table 15 reports segment means. Table 16 reports actual customer-cluster adjusted contrast intervals: First Class minus Same Day is .2528 [.2454, .2602], and Africa minus LATAM is -.0283 [-.0388, -.0178]. Both exclude zero after their familywise corrections. Equal-means H04 is therefore inconsistent with this interim validation evidence in both families; this is conditional exploratory evidence, not final-test or causal confirmation.

Within-mode validation AUC is .878 for First Class, .545 for Same Day, .546 for Second Class, and .519 for Standard Class. Pooled ranking conceals weak ordering within several modes. At threshold .50, almost no positive Standard Class orders are identified. Review workload and missed cases require segment-level assessment; explanations must be rechecked if the final model changes.

Table 15

Logistic explanations and ranking by segment

Segment	n	Mode share	Clustered 95% CI	AUC
First Class	2,021	0.881	0.879 to 0.882	0.878
Same Day	702	0.628	0.623 to 0.633	0.545
Second Class	2,546	0.695	0.693 to 0.698	0.546
Standard Class	7,882	0.714	0.713 to 0.715	0.519
Africa	710	0.714	0.707 to 0.720	0.757
Europe	3,701	0.731	0.728 to 0.734	0.743
LATAM	3,453	0.742	0.739 to 0.745	0.736
Pacific Asia	3,540	0.727	0.724 to 0.730	0.739
USCA	1,747	0.728	0.723 to 0.732	0.732

Note. Mode share is the mean normalized absolute shipping-mode contribution on the expanded logistic log-odds scale. Pointwise intervals are customer-cluster robust. AUC is computed within each segment, so between-mode separation cannot inflate within-mode ranking.

Pointwise mean intervals are not tests of differences. The actual adjusted contrast intervals and H04 interpretation appear in Table 16. These quantify model explanations, not intervention effects.

Table 16

Customer-cluster adjusted RQ4 contrasts

Comparison (A minus B)	Difference	Adjusted interval	Zero excluded?
First Class - Same Day	0.2528	0.2454 to 0.2602	Yes
First Class - Second Class	0.1855	0.1815 to 0.1896	Yes
First Class - Standard Class	0.1669	0.1640 to 0.1697	Yes
Same Day - Second Class	-0.0672	-0.0752 to -0.0593	Yes
Same Day - Standard Class	-0.0859	-0.0933 to -0.0785	Yes
Second Class - Standard Class	-0.0187	-0.0226 to -0.0147	Yes
Africa - LATAM	-0.0283	-0.0388 to -0.0178	Yes
Europe - Pacific Asia	0.0043	-0.0018 to 0.0105	No
LATAM - Pacific Asia	0.0152	0.0090 to 0.0213	Yes

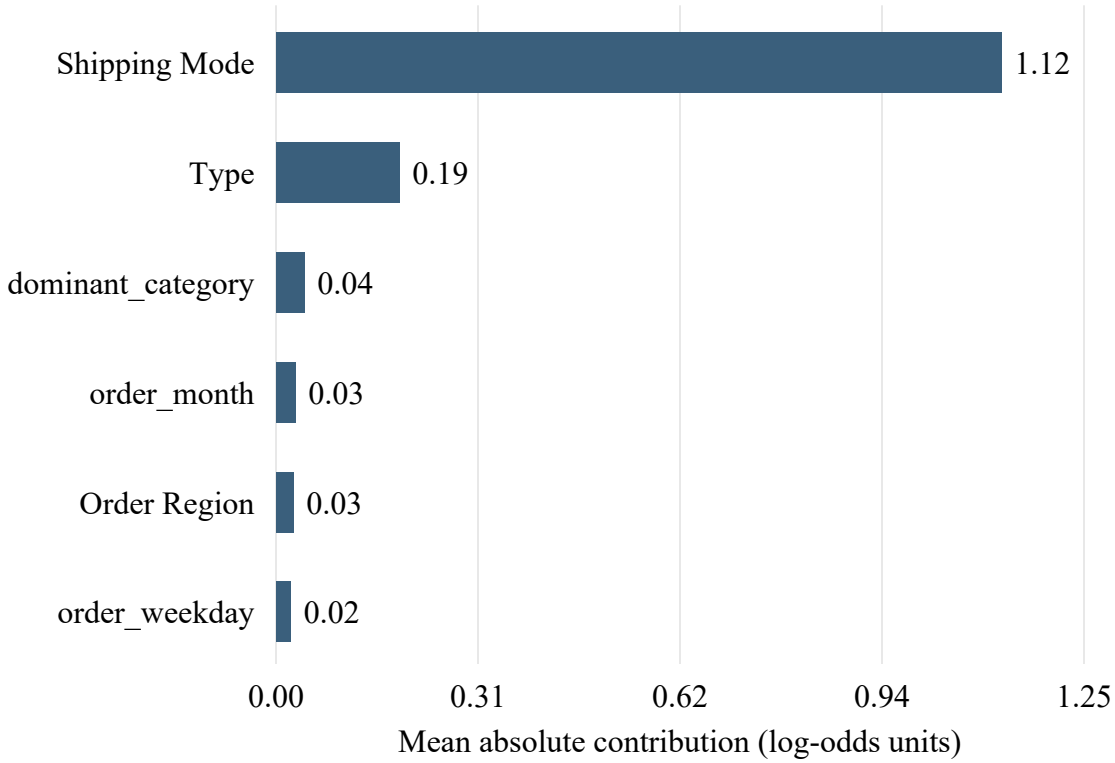
Note. All six shipping comparisons use Bonferroni familywise alpha .05 (99.167% intervals). The three displayed market comparisons are selected from all ten, each with 99.5% intervals. The market/ship families are controlled separately. Full ten market contrasts are in segment_contrasts.csv.

Conclusion for H04

At the interim validation stage, equal segment means are inconsistent with the adjusted contrasts in both shipping-mode and market families. First Class versus Same Day differs by .2528; Africa versus LATAM differs by -.0283. Europe versus Pacific Asia spans zero. Statistical significance of the small market contrasts does not establish operational importance. This conclusion is conditional on the expanded logistic model and is not a final-test or causal finding.

Figure 5

Largest mean absolute grouped logistic contributions



Note. Contributions use training-mean background values; one-hot indicators are grouped by original variable. These explain the expanded logistic model only. Correlated predictors limit causal interpretation.

Practical implication

Service category dominates the score, yet within-mode AUC is near chance for Same Day, Second Class, and Standard Class. The current model is better at separating service groups than prioritizing individual orders inside several groups. Confirm label semantics and compare service-based review rules before considering operational adoption.

Preliminary results

Interim Limitations and Risks

The development audit establishes the outcome arithmetic: non-canceled orders are labeled late when actual shipping days exceed scheduled days; canceled orders are non-late even when that duration comparison is positive. Thus the label reflects service promises and cancellation treatment. Actual customer arrival timestamps, carrier context, weather, staffing, and intervention outcomes are unavailable. Historical prediction of this indicator does not directly validate current physical arrival-delay risk.

One record per order removes repeated line labels, but customers can place multiple orders. Customer-cluster inference and resampling address some dependence in standard errors. They do not remove training-validation customer overlap, unmeasured confounding, temporal selection, or dependence across shared products and routes. The sample-size approximations assume independent-order information; design-effect and variance sensitivities show when the available sample would be inadequate.

The model recipes are initial configurations without extensive tuning. Validation outcomes have already informed the project, so further decisions can overfit this development sample. Conditional bootstrap intervals omit fitting and selection variability. The inferential model assumes additive numeric log odds and a pooled service contrast. The explanation model uses an interventional background and cannot identify causal effects among correlated predictors.

Operational inputs are incomplete. No staffing capacity, cost of a missed late order, cost of a false alert, or intervention effectiveness is available. At the reference threshold, expanded logistic flags 35.56% of orders while missing 45.34% of positive labels. The reproducibility package is publicly accessible and its principal links were verified without sign-in. Reproduction still requires downloading the attributed raw source and installing the recorded dependencies; access alone does not establish transportability or operational value.

Preliminary results

Next Steps (for Final Report)

- Maintain the public data/code snapshot and document changes. Reproduce the source audit, outcome reconciliation, models, variance pilot, and 5,000-draw paired bootstrap before the final report.
- Freeze a limited training-only tuning and calibration protocol. Retain constant, mode-lookup, and restricted-logistic benchmarks. Check numeric functional forms, the four-level service contrast, customer-separated performance, and explanation stability.
- Agree review capacity and error costs with stakeholders before selecting an operating threshold. Table 17 shows the validation trade-off: a 20% review budget captures 33.20% of positive labels at 90.99% precision. This is a scenario, not a validated intervention or savings estimate.
- Freeze features, model, calibration, threshold, comparison endpoints, and explanations before scoring the 13,151 reserved test orders. Report uncertainty and segment behavior. Audit service promises and label semantics, then use a controlled pilot before claiming fewer late deliveries or financial benefits.

Table 17

Validation scenarios for a ranked review queue

Review fraction	Alerts	Precision	Recall
10%	1,316	100.00%	18.25%
20%	2,631	90.99%	33.20%
35%	4,603	84.92%	54.22%
50%	6,576	71.81%	65.49%

Note. Top-ranked validation orders; stable tie ordering. Scores and budgets have not been validated as an operational policy. The observed 100% yield at the smallest budget is sample-specific and must not be promised for future orders.

Table 18

Interim answer and remaining work by research question

RQ	Current answer	Next required check
RQ1	Strong adjusted service association; small numeric associations uncertain	Four-level and functional-form sensitivity
RQ2	AUC about .74; most signal from mode	Frozen independent-test evaluation
RQ3	No clear gain from greater complexity	Training-only tuning and final paired comparison
RQ4	Adjusted contrasts contradict equal segment means in both families	Conditional on logistic model; check final explanation stability

Bibliography

Baryannis, G., Dani, S., & Antoniou, G. (2019). Predicting supply chain risks using machine learning: The trade-off between performance and interpretability. *Future Generation Computer Systems, 101*, 993-1004. https://doi.org/10.1016/j.future.2019.07.059

Breiman, L. (2001). Random forests. *Machine Learning, 45*(1), 5-32. https://doi.org/10.1023/A:1010933404324

Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review, 78*(1), 1-3. https://doi.org/10.1175/1520-0493(1950)078%3C0001:VOFEIT%3E2.0.CO;2

Choi, T.-M., Wallace, S. W., & Wang, Y. (2018). Big data analytics in operations management. *Production and Operations Management, 27*(10), 1868-1883. https://doi.org/10.1111/poms.12838

Christopher, M., & Peck, H. (2004). Building the resilient supply chain. *The International Journal of Logistics Management, 15*(2), 1-14. https://doi.org/10.1108/09574090410700275

Bibliography continued

Constante, F., Silva, F., & Pereira, A. (2019). *DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS* (Version 5) [Data set]. Mendeley Data. https://doi.org/10.17632/8gx2fvg2k6.5

DeLong, E. R., DeLong, D. M., & Clarke-Pearson, D. L. (1988). Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach. *Biometrics, 44*(3), 837-845. https://doi.org/10.2307/2531595

Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *The Annals of Statistics, 7*(1), 1-26. https://doi.org/10.1214/aos/1176344552

Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *The Annals of Statistics, 29*(5), 1189-1232. https://doi.org/10.1214/aos/1013203451

Hanley, J. A., & McNeil, B. J. (1982). The meaning and use of the area under a receiver operating characteristic (ROC) curve. *Radiology, 143*(1), 29-36. https://doi.org/10.1148/radiology.143.1.7063747

Bibliography continued

Ho, W., Zheng, T., Yildiz, H., & Talluri, S. (2015). Supply chain risk management: A literature review. *International Journal of Production Research, 53*(16), 5031-5069. https://doi.org/10.1080/00207543.2015.1030467

İlbaş, İ. N., Paksoy, A., & Altay, Y. (2026). A three-stage machine learning framework for late delivery risk classification, delay prediction, and explainability in supply chain management. In *Proceedings of the 9th European International Conference on Industrial Engineering and Operations Management* (Paper 27). IEOM Society International. https://doi.org/10.46254/EU09.20260027

Ivanov, D., Dolgui, A., & Sokolov, B. (2019). The impact of digital technology and Industry 4.0 on the ripple effect and supply chain risk analytics. *International Journal of Production Research, 57*(3), 829-846. https://doi.org/10.1080/00207543.2018.1488086

Liang, K.-Y., & Zeger, S. L. (1986). Longitudinal data analysis using generalized linear models. *Biometrika, 73*(1), 13-22. https://doi.org/10.1093/biomet/73.1.13

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in neural information processing systems* (Vol. 30, pp. 4765-4774). https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html

Bibliography continued

Tang, C. S. (2006). Perspectives in supply chain risk management. *International Journal of Production Economics, 103*(2), 451-488. https://doi.org/10.1016/j.ijpe.2005.12.006

Thomas, A., & Panicker, V. V. (2023). Application of machine learning algorithms for order delivery delay prediction in supply chain disruption management. In B. B. V. L. Deepak, M. V. A. R. Bahubalendruni, D. R. K. Parhi, & B. B. Biswal (Eds.), *Intelligent manufacturing systems in Industry 4.0* (pp. 491-500). Springer. https://doi.org/10.1007/978-981-99-1665-8_42

Toorajipour, R., Sohrabpour, V., Nazarpour, A., Oghazi, P., & Fischl, M. (2021). Artificial intelligence in supply chain management: A systematic literature review. *Journal of Business Research, 122*, 502-517. https://doi.org/10.1016/j.jbusres.2020.09.009

Waller, M. A., & Fawcett, S. E. (2013). Data science, predictive analytics, and big data: A revolution that will transform supply chain design and management. *Journal of Business Logistics, 34*(2), 77-84. https://doi.org/10.1111/jbl.12010

Yang, M., Lim, M. K., Qu, Y., Ni, D., & Xiao, Z. (2023). Supply chain risk management with machine learning technology: A literature review and future research directions. *Computers & Industrial Engineering, 175*, 108859. https://doi.org/10.1016/j.cie.2022.108859

Appendix A

Reproducibility and Evaluator Access

Obtain DataCoSupplyChainDataset.csv and DescriptionDataCoSupplyChain.csv from the attributed Version 5 publisher record. Read the structured file with Latin-1 encoding. Do not place raw identifying records in a public repository.

Verify structured-file SHA-256:
fa6d022ed437155e1a2f0378710602848703c8a7f203f7ff5d77805bf8480aa6

Run baseline_audit.py --data /path/to/DataCoSupplyChainDataset.csv. Then run enhance_analysis.py /path/to/DataCoSupplyChainDataset.csv from the same package folder. Then run pilot_planning.py and audit_outcome_and_bootstrap.py /path/to/DataCoSupplyChainDataset.csv. These four scripts recreate aggregation, models, inference, explanations, variance planning, outcome reconciliation and paired uncertainty. Outputs are written beside the scripts.

Install the pinned versions in requirements.txt in a clean Python environment. python 3.12.14; numpy 2.3.5; pandas 3.0.1; scipy 1.18.1; scikit-learn 1.9.1.

The 60/20/20 order split uses seed 640 and within-class floor allocations at 60% and 80%. Order-bootstrap seed is 642 (1,000 draws); pointwise customer-bootstrap seed is 643 (500 draws). Primary paired-loss intervals use 5,000 customer draws, seed 645. The training-only variance pilot uses seed 644.

The source file is not included. Raw order/customer identifiers, dates and status are used locally to regenerate clustered and temporal checks; these are not exported as public feature records. Processed rows carry only non-identifying analytical variables and partition membership. Reserved-test prediction metrics are never computed.

Appendix A continued

Table A1

Published analysis package contents

Path	Purpose
baseline_audit.py; enhance_analysis.py	Executed audit and extended analysis
audit_outcome_and_bootstrap.py	Outcome audit and 5,000-draw paired intervals
pilot_planning.py; pilot_planning_results.json	Training-only out-of-fold variance planning
processed_orders.csv	65,752 de-identified feature rows; partition labels
enhanced_validation_predictions.csv	Validation labels and six probability columns
analysis_results.json; enhanced_results.json	Audit, descriptive, fitted and uncertainty results
adjusted_associations.csv	All adjusted coefficients and clustered intervals
segment_results.csv; segment_contrasts.csv	Segment metrics and multiplicity-adjusted contrasts
logistic_grouped_explanations.csv	Mean absolute grouped contributions
raw_data_dictionary.csv; requirements.txt; README.md	Source dictionary, dependencies and execution instructions

Verified repository access

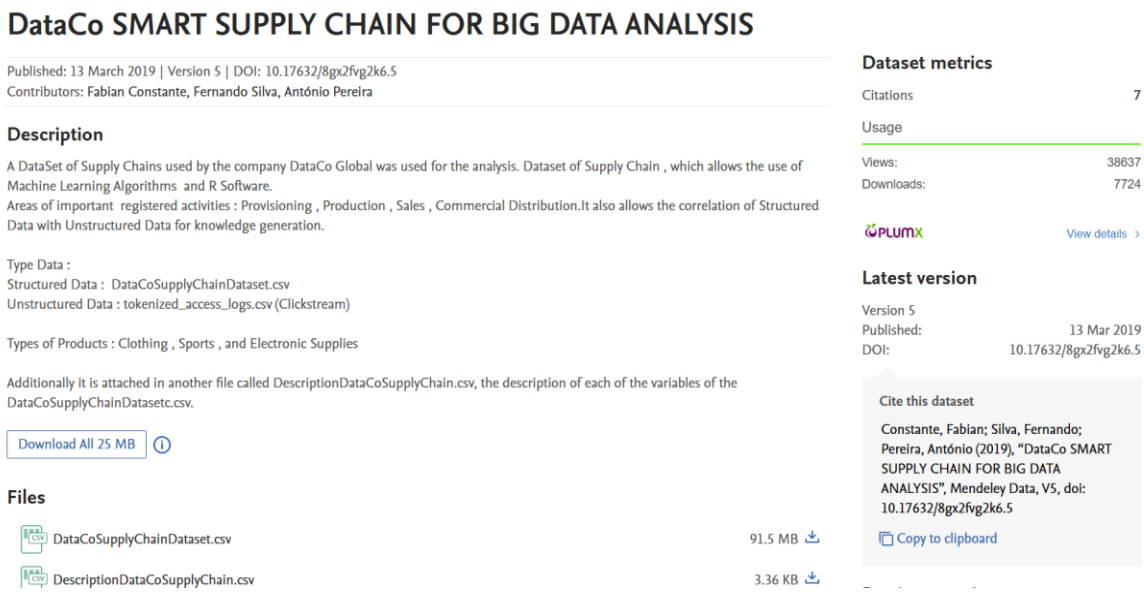
Published folder: interim_2026_10_27/. Its README links the processed data, source-audit outputs, model results, and four execution scripts. Anonymous downloads of the cleaned CSV, analysis code, pilot results, and README returned HTTP 200 and matched the committed files. The data/code snapshot is 1bda9d8; raw source access remains the publisher URL.

Label note: the sensitivity excludes the source category "Shipping canceled". Remaining rows are non-canceled source records; this is not independent verification that every order physically completed delivery.

Appendix B

Figure B1

Publisher source record screenshot



Note. Source screenshot retained from October 13, 2026; it identifies Version 5 and the structured/description files. From Constante et al. (2019), Mendeley Data. CC BY 4.0. Usage counters are historical.

Figure B2

Public repository tree snapshot

```
qm640-late-delivery-capstone/
  data/raw/README.md; scripts/; outputs/; README.md
  interim_2026_10_27/
    processed_orders.csv; raw_data_dictionary.csv
    baseline_audit.py; enhance_analysis.py; pilot_planning.py
    enhanced_results.json; pilot_planning_results.json
    segment_results.csv; segment_contrasts.csv; README.md
```

Note. Rendered from the public GitHub tree verified October 27, 2026, at data/code commit 1bda9d8. This records accessible files rather than claiming a browser screenshot. The processed dataset preview is Figure 1.